The Interstellar Permafrost Hypothesis for 3I/ATLAS (C/2025 N1)

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The interstellar object 3I/ATLAS (C/2025 N1) exhibits several anomalous characteristics: an extreme CO_2/H_2O ratio (8:1), unprecedented negative dust polarization (-2.7%), hyper-steep brightening prior to perihelion ($n \approx 7.5$), early high Ni/Fe ratio subsequenly diluted, and, perhaps its most surprising feature, a persistent sunward anti-tail. We propose a testable natural model for 3I ATLAS that would explain the different properties observed. The nature of 3I could be explained by its 8–11 Gyr of galactic cosmic ray (GCR) processing during interstellar transit, producing a 15–20 meter radiation-hardened organic crust. The observed 3 AU activation represents thermal breach of this insulating mantle, triggering pressurized CO_2 release through collimated fractures. Large momentum-dominated dust aggregates (> 1 μm) naturally explain both the polarization signature and anti-tail geometry. We provide three testable predictions for post-perihelion observations, including JWST/MIRI silicate feature suppression, thermal lag effects, and sequential metal freeze-out.

I. INTRODUCTION

The detection of the third interstellar object (ISO), 3I/ATLAS (C/2025 N1) [1, 2], has provided a rare opportunity to characterize a relic from the early Galaxy. Preliminary multi-wavelength observations indicate that 3I/ATLAS exhibits significant physical and chemical deviations from both Solar System comets and previously identified ISOs [3–8].

Kinematic analyses suggest 3I/ATLAS is an ancient object, likely originating 7–13 Gyr ago during the Milky Way's era of peak star formation [1]. While its specific galactic origin (thin vs. thick disk) remains a subject of active debate [9, 10], its residence time in the interstellar medium (ISM) implies prolonged exposure to the interstellar radiation field and galactic cosmic rays (GCR).

Spectroscopic characterization by JWST and SPHEREx revealed an unprecedented CO_2/H_2O ratio of ∼8:1 [11]. This extreme carbon enrichment, relative to the typical ratios found in Oort Cloud comets, suggests the detection of radiation-processed volatile layers rather than a primordial surface. Laboratory studies confirm that GCR irradiation of CO-rich ices synthesizes CO_2 and complex organic residues [12], leading to estimates of a radiation-hardened insulating crust approximately 15–20 m deep [13].

Thermophysical models of the object's light curve further constrain its surface properties. The activation of outgassing was characterized by a hyper-steep brightening index ($r^{-7.5\pm1.0}$) near 3 AU [14]. This threshold behavior, coupled with a low albedo ($A_v < 0.2$) and a significant thermal lag between peak solar heating and maximum activity, suggests substantial thermal inertia and an insulating mantle [15, 16].

Morphological anomalies include a persistent sunward anti-tail [17] and an "unprecedentedly deep" negative polarization of -2.7% [18]. High-resolution imaging from the Hubble Space Telescope (January 2026) further revealed the emergence of new, highly collimated jets, indicating localized structural breaches in the nucleus surface.

In this Research Note, we introduce the **interstellar permafrost hypothesis**. We argue that these disparate observations—chemical, photometric, and morphological—can be unified by a single physical model of a radiation-processed, thermally-layered nucleus. In the following sections, we detail the dust grain physics, the organometallic chemistry, and the thermal diffusion processes that define this model, concluding with three testable predictions for the outbound trajectory of 3I/ATLAS.

II. DUST PROPERTIES AND THE POLARIZATION ANOMALY

The polarimetric signature of 3I/ATLAS, characterized by a deep negative branch ($P_{min} \approx -2.7\%$) at small phase angles [18], serves as a primary constraint on the coma's grain size distribution and composition. This value deviates significantly from the standard -1% to -1.5% observed in most Oort Cloud comets [19], instead resembling the signatures of primitive Trans-Neptunian Objects (TNOs) and D-type asteroids [20].

We propose that this indicates a population dominated by large, fluffy, organic-rich aggregates, a natural consequence of its low-metallicity origin and ISM residence.

III. THE CARBONYL MECHANISM AND THERMAL STRATIFICATION

Next, we address the unique chemical evolution of the coma. One of the most significant early puzzles of 3I/ATLAS was the extreme Nickel-to-Iron (Ni/Fe) ratio detected at large heliocentric distances. While initial interpretations suggested a non-natural metal alloy [21],

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we propose this is a signature of differential sublimation within the permafrost layers.

A. Sequential Sublimation of Organometallics

In a thermally layered nucleus, heat penetrates the insulating crust over time. We argue that the Ni/Fe ratio is not a reflection of bulk elemental abundance, but rather the sequential release of volatile organometallic precursors synthesized during the object's GCR-processed history:

Nickel Tetracarbonyl [$Ni(CO)_4$]: Highly volatile, with a sublimation temperature triggering activity beyond 3 AU. This accounts for the early appearance of atomic Nickel in the coma.

Iron Pentacarbonyl [$Fe(CO)_5$]: Less volatile, remaining sequestered in the permafrost until the thermal wave reaches deeper layers or the object crosses the ∼2.6 AU threshold.

The "normalization" of the Ni/Fe ratio to standard cometary values [22] as 3I approached perihelion is a natural consequence of this sequential heating. This process confirms the stratified nature of the nucleus and would validate the high thermal inertia of the insulating crust.

B. Anti-tail Formation and Delayed Activation

The "blueing" of the coma [14] and the persistent sunward anti-tail [17] are coupled phenomena in the permafrost model. The blue shift at perihelion represents the sudden dominance of C_2 Swan bands and CN emission, pumped by solar UV resonance fluorescence, following the pressurized breach of the CO_2-rich subsurface layer.

The anti-tail consists of large ($> 1\mu$m) silicate grains with organic mantles. Unlike fine dust, these particles are momentum-dominated. Ejected through collimated fractures (nozzles) in the rigid permafrost, they remain on their original orbital trajectories, appearing sunward to observers. The persistence of the anti-tail over 100,000 km is due to the refractory nature of these grains, which survive long after their volatile components have sublimated.

IV. SECTION 4: TESTABLE PREDICTIONS FOR THE OUTBOUND PHASE

The validity of the permafrost model rests on its ability to predict the behavior of 3I/ATLAS as it recedes from perihelion. We propose three diagnostic observations to be conducted via JWST and ground-based assets through mid-2026.

a. ***Silicate Feature Suppression (JWST/MIRI).*** *The presence of large, porous aggregates and organic-rich mantles should manifest in the mid-infrared. While standard comets show a sharp, high-contrast emission peak at 10 μm due to sub-micron crystalline silicates, our model predicts a flattened or plateau-like silicate feature. This suppression occurs because the large size of the aggregates ($> 1\ \mu m$) relative to the emission wavelength makes them more efficient blackbody emitters, masking the sharp vibrational resonances of the constituent silicate grains.*

b. ***Thermal Lag and Sustained Activity.*** *If the 15–20 m crust indeed possesses high thermal inertia, the nucleus will act as a "thermal battery," retaining heat well after the sub-solar point temperature begins to drop. We predict that outgassing activity will persist at larger heliocentric distances on the outbound leg compared to the inbound leg. A "slow sunset" of activity beyond 4–5 AU would confirm the insulating properties of the GCR-processed mantle.*

c. ***Sequential Metal Freeze-out.*** *Just as the appearance of metals was staggered by volatility (Nickel before Iron), we predict a sequential freeze-out. As the thermal wave recedes, Iron pentacarbonyl emissions should diminish first, while Nickel tetracarbonyl—owing to its higher volatility—should remain detectable for a longer duration.*

V. COMPARISON: STANDARD COMET VS. INTERSTELLAR PERMAFROST

VI. CONCLUSION

3I/ATLAS represents a radiation-processed relic from the early Milky Way. Its anomalous properties, while extreme, are consistent with GCR chemistry, thermal diffusion, organometallic sublimation, and classical orbital mechanics. The permafrost hypothesis is falsifiable via silicate feature suppression, thermal lag, and sequential freeze-out, enabling rapid testing in 2026.

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TABLE I. Comparative properties of Solar System comets and 3I/ATLAS under the Permafrost Model.

Characteristic	Solar System Comet	3I/ATLAS (Permafrost)
Age	$\sim$4.5 Gyr	8–11 Gyr
Origin	Solar nebula, solar metallicity	Early Galaxy, low metallicity
Surface	Thin dust mantle (<1 m)	Thick GCR-processed crust (15–20 m)
Dominant Volatile	H_2O	CO_2 (8× H_2O)
Dust Properties	Fine silicates (<1 μm)	Large porous aggregates (1–100 μm)
Activity Onset	Gradual	Sudden threshold at $\sim$3 AU
Polarization	Standard negative branch	Deep negative branch (-2.7%)
Light Curve Index	Moderate (r^{-3} to r^{-4})	Hyper-steep ($r^{-7.5}$)
Outgassing Pattern	Broad, diffuse	Collimated jets + diffuse component
Anti-Tail	Rare, transient	Persistent, sunward, $> 10^5$ km

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